EFFECTS OF FOOD INFLATION AND NON-FOOD INFLATION AGRICULTURAL OUTPUT IN GHANA

BY

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A DISSERTATION SUBMITTED TO THE DEPARTMENT OF AGRICULTURAL ECONOMICS AND AGRIBUSINESS, COLLEGE OF BASIC AND APPLIED SCIENCES, UNIVERSITY OF GHANA, LEGON, IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF SCIENCE (HONOURS) DEGREE IN AGRICULTURE. (AGRICULTURAL ECONMICS)

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DECLARATION

I, SALIM IBRAHIM, author of this dissertation, do hereby declare that the work presented in this dissertation, “EFFECTS OF FOOD INFLATION AND NON-FOOD INFLATION ON AGRICULTURAL OUTPUT IN GHANA”, was done entirely by me in the Department of Agricultural Economics and Agribusiness, University of Ghana, Legon.

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DEDICATION

This work is dedicated to the Almighty God, whose grace and wisdom have sustained me throughout this academic journey. I also dedicate this project to my loving parents and siblings for their unwavering support, and to every Ghanaian farmer who continues to work tirelessly under uncertain economic conditions to feed the nation.

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ABSTRACT

This study examines the effect of food and non-food inflation on agricultural output in Ghana using annual time series data spanning from 1991 to 2022. Employing the Autoregressive Distributed Lag (ARDL) bounds testing approach, the research investigates both the long-run and short-run relationships between inflation categories and agricultural productivity, while accounting for exchange rate, input cost, GDP per capita, and credit to the agricultural sector.

Trend analysis reveals an upward trend in agricultural output over the study period, while both food and non-food inflation exhibit downward trends. The ARDL results confirm a long-run relationship between agricultural output and the explanatory variables. In the long run, food inflation negatively affects agricultural output, while GDP per capita and credit to agriculture exert a positive and statistically significant influence. Interestingly, the cost of agricultural inputs shows a positive effect contrary to prior expectations, potentially reflecting increased investment behaviour among commercial farmers.

Short-run dynamics show that lagged agricultural output and inflation variables significantly influence current output. Diagnostic tests confirm the robustness of the model, including normality, no autocorrelation, homoscedasticity, and correct model specification. The stability of the model was confirmed using the CUSUM test.

The study concludes that inflation especially food inflation has a constraining effect on agricultural growth in Ghana. Policy recommendations include enhancing price stability, improving access to credit, and targeted investment in agricultural inputs to ensure sustainable agricultural productivity growth.

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LIST OF ABBREVIATIONS

ARDL	Autoregressive Distributed Lag
ECM	Error Correction Model
AIC	Akaike Information Criterion
FAO	Food and Agriculture Organization
GSS	Ghana Statistical Service
ISSER	Institute of Statistical, Social and Economic Research
CUSUM	Cumulative Sum
CUSUMSQ	Cumulative Sum of Squares
CPI	Consumer Price Index
GDP	Gross Domestic Product
WDI	World Development Indicators

CHAPTER ONE

INTRODUCTION

1.1. Background

Ghana's agricultural sector has been a significant contributor to the country's economy, employing about 47% of the economically active population and accounting for approximately 30% of GDP over the past three decades (Nyamekye et al., 2021; E. Obeng, 2023). However, its contribution has declined from 35.2% to about 19% in recent years due to challenges such as climate change and limited financial credit (Céu & Gaspar, 2024). The central role of agriculture in Ghana's economy indicates that increase in output of the sector could trigger overall growth of the economy and reduce poverty level. Maintaining high agricultural productivity is not only capable of stimulating aggregate economic growth in the country but also has the potential of reducing severe rural poverty (Mohammed et al., 2021).

The importance of agriculture to national development lies in its potential to stimulate broad-based economic growth. Empirical evidence from other developing economies supports the notion that increased agricultural output contributes positively to GDP growth, reduces food insecurity, and drives employment, particularly in rural areas (Poudel et al., 2021; Rafael, 2023). In Several studies have noted that improvements in agricultural research, extension services, and input supply chains have yielded productivity gains in recent years (Hazrana & Mishra, 2024). Nevertheless, structural bottlenecks such as insufficient public investment persist. Ghana's government, for instance, allocated only 1.3% of the national budget to agriculture in 2004 far below the 10% commitment outlined in the 2003 Maputo Declaration (Islamic Development Bank Institute et al., 2023). The underfunding of the sector is widely seen as a contributing factor to stagnant output and limited modernization.

Moreover, the economic transition following the 2007 oil discovery has accelerated the reorientation of Ghana's economy toward services and extractive industries. As a result, agriculture has lost its status as the dominant contributor to GDP, despite its continued importance in employment and rural livelihoods (Kunkuaboor et al., 2021).

In parallel, inflation has emerged as a major macroeconomic challenge affecting the performance of key economic sectors, including agriculture. Abdullah et al., (2024) also found that inflation in Ghana is influenced by multiple drivers, including exchange rate volatility, fiscal deficits, supply-side constraints, and global commodity prices. According to Boamah (2019), budget deficits and public debt accumulation have become structural sources of inflationary pressure, aligning with the fiscal theory of inflation which posits that governments may resort to inflationary financing to cover fiscal imbalances (Seidu & Vasilev, 2024). In the short run, studies have shown that import prices and real income per capita are key inflation drivers, while money supply plays a limited role (Boamah, 2019).

Given these dynamics, understanding how food and non-food inflation affect agricultural productivity is critical for designing responsive policies. Such policies must balance inflation control with targeted support for the agricultural sector through mechanisms like concessional credit, input subsidies, and trade policy interventions. As Abdu et al., (2018) emphasized, price stability, especially for food items, must be prioritized alongside broader macroeconomic reforms to ensure agricultural sustainability and inclusive growth.

1.2. Problem statement

The effect of inflation on agricultural output in Ghana is multifaceted and significant. Ghana's agricultural sector, encompassing farming, forestry, and fishing, plays a pivotal role in the economy, employing nearly half of the workforce and contributing substantially to GDP. Despite its importance, the sector remains highly vulnerable to inflation (Boateng et al., 2022).

While existing studies have explored the relationship between inflation and agricultural output globally, there is limited empirical research addressing the Ghanaian context. Specifically, the literature does not adequately examine the differential effects of food inflation and non-food inflation on agricultural output. This gap is critical, as this study seek to fill this knowledge gap by exploring the long-run and short-run relationship between agricultural output, food and non-food inflation, and other macroeconomic variables. Key challenges include:

1. Non-food inflation raises the cost of essential agricultural inputs, making it challenging for farmers to maintain or increase their production levels, ultimately leading to reduced output (Alta et al., 2021).
2. Food inflation can lead to higher prices for agricultural produce, which may boost farmer productivity but also increase food costs for consumers (Mbah et al., 2022).

This study is aimed at analyzing the differential effects of inflation on agricultural output by providing answers to the following questions:

1.What has been the trend in inflation (food and non-food) and agricultural output in Ghana from 1991 to 2022

2. How does food inflation and non-food inflation influence agricultural output in Ghana?

1.3. Research objectives

The primary objective of this study is to examine the effect of inflation on agricultural output in Ghana. The specific objectives are;

1. Describe the trend in agricultural output, food inflation and non-food inflation in Ghana from 1991 to 2022.
2. To examine the magnitude and effects of food inflation and non-food inflation on agricultural output in Ghana.

1.4. Justification of the study

Understanding the differential effect of food inflation and non-food inflation on agricultural output is essential for tailoring policies to the unique structure of Ghana's agricultural economy. There is also limited empirical research that specifically quantifies the effect of inflation on agricultural output in Ghana, this gap in knowledge hinders the development of targeted policies to mitigate the adverse effects of inflation on the agricultural sector. Addressing this gap could improve decision-making on monetary and fiscal policies to mitigate inflation's adverse effects while promoting sustainable agricultural growth and food security. The analysis of the effect of inflation on agricultural output in Ghana will improve policy design through targeted interventions, monetary policy adjustments and fiscal policy adjustments. The study will also add to the existing literature and provide current analysis of the effect of inflation on agricultural output in Ghana. The study would also be a good source of information to both the private sector and the public sector (particularly the Economic Management Team) that use inflation as an important input in their decisions. The study would also provide additional information on inflation in an attempt to fight high and unstable inflation and pave way for agricultural growth and productivity.

1.5. Organization of the Study

This study will be organized into five chapters. The first chapter comprises the background, problem statement, objectives of the research, relevance of the research and the organization of the research. Chapter two reviews the literature related to the study. The third chapter discusses the methodology that will be used to achieve the objectives of the study. Chapter four presents the results and discussions of the study. The study then concludes the with chapter five that consist of the summary, conclusions and recommendations of the study.

CHAPTER TWO

LITERATURE REVIEW

2.1. Introduction

This chapter provides a review of relevant literature concerning the differential effect of inflation and macroeconomic variables on agricultural output. It synthesizes empirical findings that relate to the study objectives. The review is structured thematically to reflect the variables to be examined, including food and non-food inflation, exchange rate, total credit, and economic growth, while also integrating insights from Ghana-specific and comparative studies.

2.2. Historical trends and patterns of inflation in Ghana

Inflation in Ghana has exhibited significant fluctuations over the past several decades, reflecting shifts in macroeconomic policy, global shocks, and domestic structural challenges (Boamah, 2019). During the immediate post-independence period (1957–1964), Ghana maintained relative price stability, with inflation remaining within single-digit levels. This was largely attributed to a stable fiscal environment and modest government expenditure (Seidu & Vasilev, 2024).

However, the macroeconomic landscape shifted drastically in the 1970s and early 1980s. The country experienced runaway inflation, with rates exceeding 100% in 1977, 1981, and 1983. These hyperinflationary episodes were driven by deteriorating fiscal balances, collapse in commodity export revenues, and political instability. The period between 1975 and 1983 saw the economy contract significantly, with average annual GDP growth falling by over 2% (Seidu & Vasilev, 2024).

To counteract this instability, Ghana adopted stabilization and structural adjustment policies from 1983 onwards, supported by the Bretton Woods Institutions. These policies were implemented in phases: the Economic Recovery Programme (1983–1986), the Structural Adjustment Programme (1987–1990), and the Accelerated Growth Strategy (1991–2000) (Huq & Tribe, 2018). Although these reforms helped liberalize the economy and attract foreign aid, inflation remained relatively high throughout the period, averaging around 33% annually. Temporary drops in inflation such as in 1985 and 1992 were not sustained, suggesting that the underlying structural issues remained unresolved (Dossche, 2009; Levin & Piger, 2002). The early 2000s marked a turning point. With improved monetary policy targeting by the Bank of Ghana, introduction of inflation-targeting frameworks, and relative fiscal consolidation, inflation trends began to moderate (Abango et al., 2019). The mid-2000s saw inflation fall to low double digits, with single-digit rates recorded in some years (notably in 2011 and 2013). However, external shocks such as oil price volatility and currency depreciation periodically reignited inflationary pressures (Amoatey et al., 2024; Boamah, 2019).

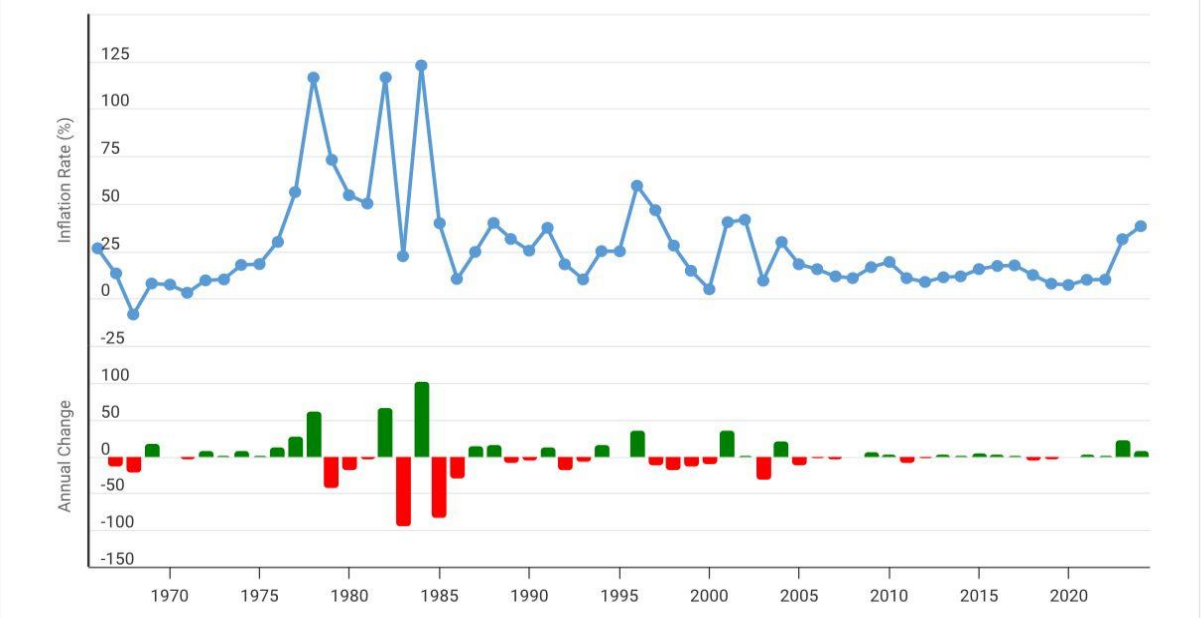


Figure 2.1. Ghana Inflation Rate (1965-2024)

Source: https://www.macrotrends.net/global-metrics/countries/GHA/ghana/inflation-rate-cpi

2.3. Empirical Review

2.3.1 Effect of Inflation on Agricultural Output

Empirical studies demonstrate mixed results on the inflation-agriculture nexus. A study by Mohammed et al. (2020) on Ghana confirmed a long-run relationship between agricultural output and economic growth, with inflation exerting a significant negative effect on GDP. The study showed that a 1% increase in inflation reduced economic growth by 0.031%, highlighting inflation's potential to undermine agricultural-driven economic progress.

Similarly, in a study focusing on Nigeria, (Catherine et al., 2024; Mbah et al., 2022)) found that inflation, along with exchange rate fluctuations, significantly constrained agricultural output. Using an ARDL framework, their findings emphasized that macroeconomic instability eroded the profitability and productivity of agricultural investments. The authors recommended that targeted inflation control and agricultural subsidies be used to buffer the sector against external shocks.

A threshold analysis by Tenaw & Demeke, (2020) in Ethiopia introduced a non-linear dimension to the debate, revealing that inflation has asymmetric effects depending on whether it crosses certain thresholds. They found that food inflation had a more volatile and damaging effect once it surpassed a 10% threshold, while non-food inflation became harmful beyond 8%. This suggests the need for differentiated policy approaches to different types of inflation.

Furthermore, Obeng et al., (2022) focused specifically on Ghana's exchange rate pass-through to food and non-food inflation. Their non-linear ARDL model showed that depreciation of the local currency disproportionately affected food prices. This finding has significant implications for agricultural output, given the sector's reliance on imported inputs such as fertilizers, machinery, and chemicals. The study emphasized the asymmetric and stronger pass-through of

exchange rate depreciation on food inflation than appreciation, affirming that stabilizing the exchange rate is critical for managing agricultural inflation.

2.3.2 Agricultural Output and Economic Growth Nexus

A robust body of literature affirms the positive contribution of agricultural output to national economic growth, particularly in developing countries. The work of Mohammed et al. (2020) on Ghana employed Johansen cointegration and VECM analysis, finding a unidirectional causality running from agriculture to GDP. This implies that agriculture remains a driver of economic growth, and that increasing output in this sector yields positive returns for the broader economy.

This aligns with findings by Tenaw & Demeke, (2020), who found that macroeconomic variables such as inflation and capital formation had significant impacts on agricultural output, and that the sector was crucial for ensuring food security and poverty reduction. Likewise, Obe-Nwaka & Agbenyi, (2023) used a similar methodology in Nigeria and found a co-integrating relationship between agricultural output and economic growth, reinforcing the need for sectoral prioritization in economic planning.

2.4 Relationship Between Food and Non-Food Inflation, Agricultural Output, and Economic Growth

Recent studies highlight the nuanced relationship between inflation components and agricultural output, especially in Sub-Saharan African economies like Ghana. Tenaw & Demeke, (2020) provided strong evidence of threshold effects, noting that beyond a certain point, both food and non-food inflation can destabilize the economy. Food inflation, due to its immediate impact on household consumption and food producers, poses a particularly acute challenge. When prices rise beyond the incentive threshold, production costs outweigh expected profits, thereby reducing output.

In Ghana, Mbah et al., (2022) found that food inflation is more responsive to exchange rate fluctuations than non-food inflation. The agricultural sector thus experiences both the benefit of higher output prices and the burden of inflated input costs. This dual effect makes food inflation a critical variable in any policy aimed at stabilizing agricultural growth.

Mohammed et al. (2020) underscored the long-run equilibrium relationship between agricultural output and economic growth in Ghana. Their analysis, covering over five decades, showed that agricultural output positively contributes to GDP and is significantly influenced by inflation and labor supply. This suggests that managing inflation—particularly in the food sector can enhance agriculture's role in driving economic expansion.

In the case of Nigeria, Tenaw & Demeke, (2020) found that while inflation generally reduced agricultural output, credit to agriculture and capital formation could act as stabilizers. These findings reinforce the idea that inflation's impact is not isolated; it interacts with other macroeconomic variables to influence productivity.

Collectively, the literature suggests that inflation management must go beyond monetary policy to encompass exchange rate stabilization and sector-specific interventions such as input subsidies, credit schemes, and investment in rural infrastructure. Targeting food inflation specifically may yield better results than generalized anti-inflation policies. Understanding these dynamics is essential for formulating evidence-based strategies to enhance agricultural productivity and sustain economic growth in Ghana.

CHAPTER THREE

METHODOLOGY

3.1. Introduction

This chapter presents the methodological approach used to examine the differential effect of inflation on agricultural output in Ghana. It outlines the theoritical framework, conceptual framework, method of analysis, data source and as well as the study area have been employed to achieve the study's objectives.

3.2. Theoretical Framework

This study is grounded in the agricultural supply response theory, which offer insights into how macroeconomic price instability influences agricultural output, particularly in developing countries like Ghana. The agricultural supply response theory, derived from neoclassical microeconomic foundations, examines how farmers adjust their production decisions in response to changes in prices and costs. At its core, the theory assumes that rational producers aim to maximize profit and will respond to relative price changes by reallocating resources accordingly (Suh & Moss, 2021).

3.3. Conceptual Framework

The conceptual framework for this study as shown in Figure 3.1. is based on the concept that agricultural output, being the dependent variable is affected by food and no-food inflation and other macroeconomic factors such as exchange rates, input cost, total credit, and GDP per capita.

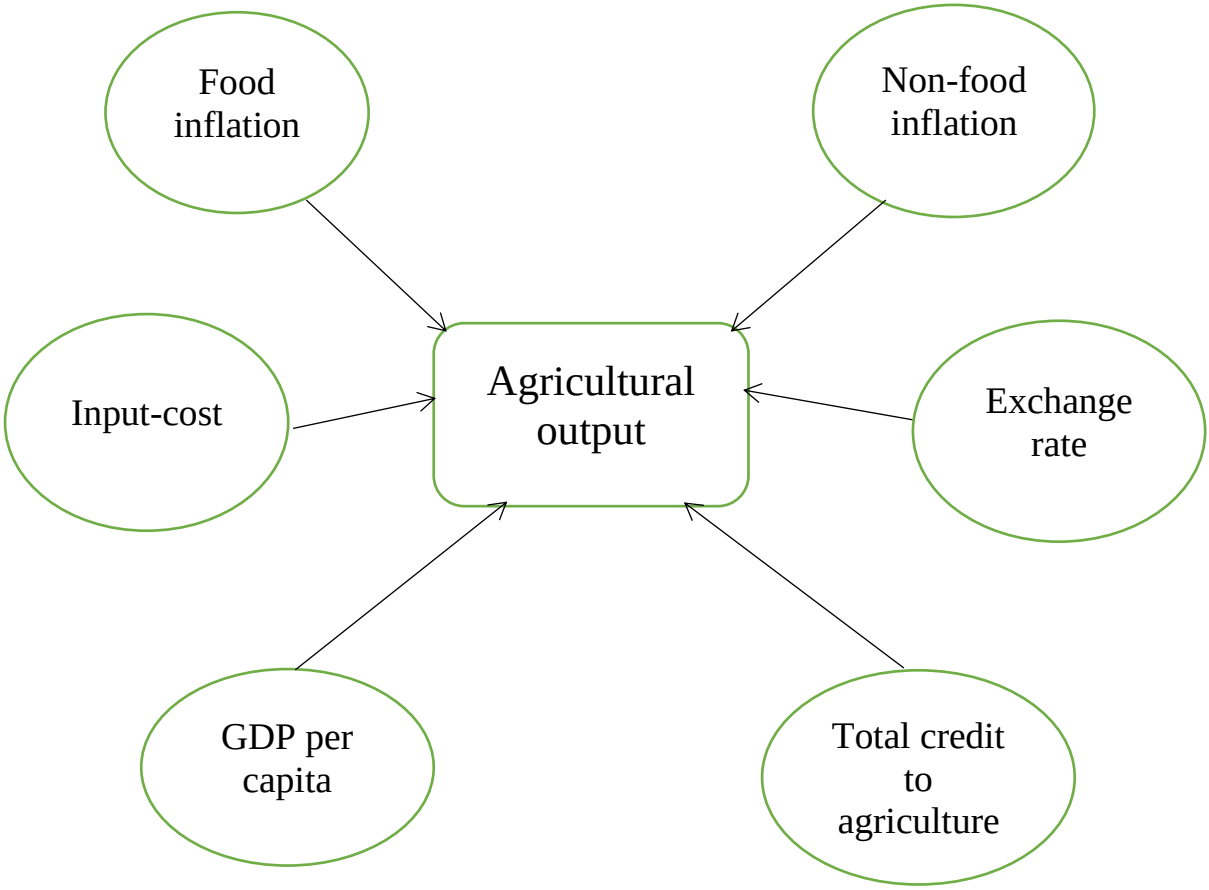


Figure 3.1. Conceptual Framework of the study

Source: Authors' construct, 2025

3.4. Data source and variable description

This study utilizes annual time series data spanning from 1991 to 2022 to examine the effect of food and non-food inflation on agricultural output in Ghana. All data employed in this analysis are secondary in nature, obtained from reliable and credible institutions.

Specifically, data on food and non-food inflation (measured in annual percentage changes) were sourced from the Ghana Statistical Service (GSS) and the Institute of Statistical, Social and Economic Research (ISSER). These inflation series represent price movements within the food and non-food components of the Consumer Price Index (CPI) basket. Agricultural output is captured using agricultural value-added, measured in U.S. dollars, 2015 prices. The data were obtained from the Food and Agriculture Organization (FAO) database. The value-added figures reflect the net output of the agricultural sector after accounting for intermediate inputs and are deflated to ensure consistency across time. Input cost is proxied by the total import value of pesticides in 1000 U.S. dollars, which serves as a representative measure of agricultural production inputs. This data was sourced from the FAO's agricultural trade indicators. Real GDP per capita, used to control for overall economic growth, was also sourced from WDI. It is calculated in constant 2015 U.S. dollars and serves as a proxy for income levels and economic well-being. Credit to the agricultural sector, measured as the total credit allocated to agriculture relative to overall credit disbursed in constant 2015 U.S. dollars terms, was similarly obtained from FAO.

By using time series data across a 32-year period, this study ensures robust statistical inference and the ability to trace dynamic relationships over time. All variables were cleaned and harmonized to maintain consistency in units and deflation base year.

3.5. Method of data analysis

3.5.1. Trend in Ghana' Agricultural Output, Food Inflation, and Non-Food Inflation.

To investigate the developmental trajectory of agricultural output, food inflation, and non-food inflation in Ghana over the study period, trend analysis will be employed. The analysis will also be supported by descriptive statistics such as the mean, standard deviation, and coefficient of variation (CV) to understand the dispersion and consistency of the variables over time. Furthermore, graphical representations will be used to observe whether each variable followed an upward or downward trend during the period under review.

In order to determine the rate at which agricultural output, food inflation, and non-food inflation have grown or declined, each of the variables will be regressed on time. The analytical model specified for this purpose was:

$$Y_t = \beta_0 + \beta_1 T + \varepsilon_t \ \ldots\ldots\ldots\ldots (1)$$

Where, Y_t represents agricultural output, food inflation, or non-food inflation at time, t. β0 is the intercept. β_1 is the slope coefficient indicating the trend direction. T is the time variable, and ε is the stochastic error term.

A positive value of B1 indicates an upward trend, while a negative value suggests a downward trend. The coefficient B1 directly reflects the average yearly change in the respective variable over the study period. This trend analysis therefore helps understand the pattern and direction of movement in agricultural output, food inflation, and non-food inflation between 1991 and 2022.

3.5.2. Magnitude and effects of food inflation and non-food inflation on Ghana's agricultural output

Linear regression model

To analyze the overall influence of macroeconomic variables on agricultural output, the study will employ the Ordinary Least Squares (OLS) method to estimate the parameters of a multiple regression model. The model is specified as follows:

$$\mathbf{AGRO_t} = \boldsymbol{\beta}_0 + \boldsymbol{\beta}_1\mathbf{F_INF_t} + \boldsymbol{\beta}_2\mathbf{NF_INF_t} + \boldsymbol{\beta}_3\mathbf{LnEXC_t} + \boldsymbol{\beta}_4\mathbf{LnINPUT_t} + \boldsymbol{\beta}_5\mathbf{LnGDPPC} + \boldsymbol{\beta}_6\mathbf{LnCREDIT} + \boldsymbol{\varepsilon_t} \ldots\ldots (2)$$

Where $\boldsymbol{\varepsilon_t}$ represents the error term. This model assumes that each macroeconomic variable independently affects agricultural output. The coefficients β1 to β6 are expected to capture the magnitude and effect of Agricultural output with respect to each explanatory variable. The natural log is applied to some variables based on economic theory and available information relating to the phenomenon being studied.

Statement of Hypotheses:

H_0 : $\boldsymbol{\beta}_i = 0$ (No significant relationship between the variable and agricultural output)

Ha : $\boldsymbol{\beta}_i \neq 0$ (There is a significant relationship)

Validation of Hypotheses:

Hypotheses are tested using the t-statistic:

$$t_c = \frac{\beta_k}{Se\ (\beta_k)}$$

Where, t_c is t-statistic, β_k is the estimated parameter of the explanatory variables and $Se\ (\beta_k)$ is the standard error of the coefficient. If the absolute value of t_c exceeds the critical value from the t-distribution at a given significance level, typically 5%, the null hypothesis is rejected.

Pre-Diagnostic and Statistical Test

Before estimating the subsequent econometric models, it is essential to examine the time series properties of the data used in the study to avoid spurious regression results.

Stationary test

Stationarity test will be conducted to determine the order of integration of each variable. This test identifies whether the statistical properties of the variables such as mean, variance, and autocorrelation are time-invariant. Non-stationary data can result in misleading inferences, hence the need to transform such series to ensure stationarity.

To this end, both Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests will be employed to determine the stationarity of the variables in both levels and first differences. The null hypothesis for both tests is that the variable has a unit root (i.e., is non-stationary). If the test statistic is more negative than the critical value, the null hypothesis is rejected in favour of the alternative that the series is stationary.

Optimal lag selection

The selection of appropriate lag length in time series models is crucial for accurate analysis and forecasting. The Akaike Information Criterion (AIC) has been found to be superior for small samples of 60 observations or fewer, minimizing underestimation and maximizing the chance of identifying the true lag length. As recommended by Wooldridge et al. (2016), a maximum lag length of 2 will be adopted to preserve degrees of freedom in annual time series. The optimal lag structure will be selected using the Akaike Information Criterion (AIC), which balances model fit and parsimony (Liew, 2021)

ARDL Bound test for cointegration

Autoregressive distributed lag (ARDL) model

This study employs the Autoregressive Distributed Lag (ARDL) bounds testing approach developed by Pesaran et al. (2001) to examine the long-run relationship between agricultural output and its selected macroeconomic determinants, including food inflation, non-food inflation, exchange rate, GDP per capita, credit to agriculture, and input cost. To formally test for cointegration, the following unrestricted ARDL model was estimated:

$$\Delta \boldsymbol{AGROt} = \boldsymbol{\beta}0 + \sum_{i=1}^{p} \boldsymbol{\beta}_1 \Delta \boldsymbol{AGRO}_{t-i} + \sum_{i=o}^{q} \boldsymbol{\delta}_1 \Delta \boldsymbol{F_INF}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_2 \Delta \boldsymbol{NF_INF}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_3 \Delta \boldsymbol{LnEXC}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_4 \Delta \boldsymbol{LnINPUT}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_5 \Delta \boldsymbol{LnGDPPC}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_6 \Delta \boldsymbol{LnCREDIT}_{t-i} + \boldsymbol{\varphi}_1 \boldsymbol{AGRO}_{t-1} + \boldsymbol{\varphi}_2 \boldsymbol{F_INF}_{t-i} + + \boldsymbol{\varphi}_3 \boldsymbol{NF_INF}_{t-i} + \boldsymbol{\varphi}_4 \boldsymbol{LnEXC}_{t-i} + \boldsymbol{\varphi}_5 \boldsymbol{LnINPUT}_{t-1} + \boldsymbol{\varphi}_6 \boldsymbol{LnGDPPC}_{t-1} + \boldsymbol{\varphi}_7 \boldsymbol{LnCREDIT}_{t-1} + \boldsymbol{\varepsilon}_t \text{ .....} (3)$$

Where, Δ denotes the first difference operator, $\boldsymbol{\beta}_0$ is the drift component or constant, p, q1, ...q6 are optimal lag lengths determined by the Akaike Information Criterion (AIC). $\boldsymbol{\varphi}_i$ are long-run parameters and εt is the stochastic error term.

The bounds testing approach to cointegration involves conducting an F-test on the joint significance of the lagged level variables in the unrestricted error correction form.

$$\boldsymbol{AGROt} = = \boldsymbol{\beta}\boldsymbol{0} + \sum_{i=1}^{p} \boldsymbol{\varphi}_1 \boldsymbol{AGRO}_{t-i} + \sum_{i=o}^{q} \boldsymbol{\varphi}_2 \boldsymbol{F_INF}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\varphi}_3 \boldsymbol{NF_INF}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\varphi}_4 \boldsymbol{LnEXC}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\varphi}_5 \boldsymbol{LnINPUT}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\varphi}_6 \boldsymbol{LnGDPPC}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\varphi}_7 \boldsymbol{LnCREDIT}_{t-i} + \boldsymbol{\varepsilon}_t \text{.....} (4)$$

Hypotheses:

The null hypothesis of no long-run relationship is defined as:

H_0: $\varphi_1 = \varphi_2 = \varphi_3 = \varphi_4 = \varphi_5 = \varphi_6 = \varphi_7 = 0$ (No long-run relationship)

versus the alternative hypothesis:

Ha: $\varphi_1 \neq \varphi_2 \neq \varphi_3 \neq \varphi_4 \neq \varphi_5 \neq \varphi_6 \neq \varphi_7 \neq 0$ (Long-run relationship exists)

The decision rule for confirming cointegration is that if the computed F-statistic exceeds the upper critical bound, the null hypothesis of no cointegration is rejected, indicating the existence of a long-run relationship between the dependent and explanatory variables.

Error correction model (ECM):

Following confirmation of a long-run cointegration relationship through the ARDL bounds test, the next step involved estimating the Error Correction Model (ECM) to capture both the short-run dynamics and the speed of adjustment to long-run equilibrium.

The ECM representation of the ARDL model is specified as:

$$\Delta \boldsymbol{AGROt} = \boldsymbol{\beta}0 + \sum_{i=1}^{p} \boldsymbol{\beta}_1 \Delta \boldsymbol{AGRO}_{t-i} + \sum_{i=o}^{q} \boldsymbol{\delta}_1 \Delta \boldsymbol{F_INF}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_2 \Delta \boldsymbol{NF_INF}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_3 \Delta \boldsymbol{LnEXC}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_4 \Delta \boldsymbol{LnINPUT}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_5 \Delta \boldsymbol{LnGDPPC}_{t-i} + \sum_{i=0}^{q} \boldsymbol{\delta}_6 \Delta \boldsymbol{LnCREDIT}_{t-i} + \boldsymbol{\gamma lnECT}_{t-1} + \boldsymbol{\varepsilon}_t \ldots\ldots\ldots\ldots (5)$$

Where, Δ donates the first difference of the variables (e.g., $\Delta AGROt = AGROt - AGRO_{t-1}$), ECT is Error Correction Term, and γ is the coefficient of the error correction term.

NB: Speed of adjustment parameter indicating how quickly deviations from the long-run equilibrium are corrected. A negative and statistically significant γ confirms adjustment towards equilibrium.

Post-Diagnostic and statistical Test

After estimating the ARDL and ECM models, several diagnostic tests were conducted to verify the validity, reliability, and robustness of the regression results.

To detect the presence of serial correlation in the residuals, the Breusch-Godfrey LM test was conducted. The null hypothesis is that there is no autocorrelation up to a specified lag order. A high p-value (greater than 0.05) implies failure to reject the null, indicating no autocorrelation. Additionally, because the model includes a lagged dependent variable, the Durbin's h-test will also be employed to reinforce the results of the LM test. The test statistic is compared to the critical value from the standard normal distribution. If the absolute h-value is less than the Z-critical value, the model is considered free from autocorrelation.

To determine whether the variance of the residuals is constant (i.e., homoscedastic), the Breusch-Pagan/Cook-Weisberg test for heteroskedasticity was employed. The null hypothesis is that the residuals have constant variance. A p-value above 0.05 leads to the acceptance of the null hypothesis, indicating the absence of heteroskedasticity and confirming that the model is homoscedastic.

The Skewness/Kurtosis test was used to examine the normality of the residuals. The null hypothesis is that the residuals are normally distributed. A p-value greater than 0.05 means the residuals follow a normal distribution, which is desirable for valid statistical inference in small samples.

To check whether the functional form of the model is correctly specified, the Ramsey RESET test was applied. The null hypothesis assumes that the model is correctly specified, and no significant omitted variables exist. A p-value above the 5% level indicates that the null hypothesis cannot be rejected, and hence, the model is deemed correctly specified.

Stability test

Beyond the standard post-estimation diagnostic tests, it is crucial to assess the stability of the model parameters to validate the reliability of the estimated relationships over the entire study period. In this study, parameter stability will be examined using the Cumulative Sum (CUSUM) and Cumulative Sum of Squares (CUSUMSQ) tests, based on the recursive residuals of the ARDL model.

These tests assess whether the estimated coefficients remain stable or whether they experience structural changes due to shifts in economic policy, external shocks, or measurement errors. The null hypothesis of the tests posits that the model's parameters are stable over time. If the plot of the cumulative statistics stays within the 5% critical bounds, then we fail to reject the null, confirming model stability. If they cross the bounds, it indicates a structural break or instability.

Variables and A Priori Expectations

This section presents the key variables used in the study along with their expected theoretical effects on agricultural output. Each variable has been carefully selected based on economic theory and empirical evidence, with particular attention to the structure of Ghana’s economy and the role of agriculture.

Agricultural Output

The dependent variable in this study is agricultural output, denoted as AGRO, and measured using value added deflator in agriculture, fishing and forestry in 2015 prices U.S. dollars. This indicator captures the net output of the agriculture sector including crops, livestock, forestry, and fishing after subtracting intermediate inputs. The use of value added provides a reliable estimate of the sector's contribution to GDP, allowing for meaningful comparisons over time. Data on agricultural value added were obtained from the Food and Agriculture Organization (FAO). A priori, agricultural output is expected to be negatively affected by non-food, exchange rate and high input costs, while macroeconomic support indicators such as food inflation, rising GDP per capita, and increased agricultural credit are anticipated to promote growth in the sector.

Food Inflation (F_INF)

Food inflation, denoted as F_INF, measures the annual percentage change in the general price level of food items. It was sourced from the Consumer Price Index (CPI) series published by the Ghana Statistical Service and Institute of Statistical, Social and Economic Research. A priori, a positive relationship is expected between food inflation and agricultural output.

Non-Food Inflation (NF_INF)

Non-food inflation (NF_INF) captures the broader macroeconomic environment by measuring the percentage increase in prices of goods and services excluding food. It reflects the cost of items such as fuel, transport, utilities, and communication, which are critical to agricultural logistics and production. Data on non-food inflation were also sourced from the Ghana Statistical Service and Institute of Statistical, Social and Economic Research. The relationship between non-food inflation and agricultural output is expected to be negative.

Exchange Rate (EXC)

The exchange rate, represented as EXC, measures the average annual nominal exchange rate of the Ghanaian cedi against the U.S. dollar. The data were collected from FAOSTAT databases. In agricultural economies reliant on imported inputs such as machinery, fertilizer, and pesticides a depreciating exchange rate (i.e., cedi weakening) increases production costs, potentially reducing output.

Thus, a negative relationship is expected between the exchange rate and agricultural output, as cedi depreciation can raise the cost of essential imports and limit farmers' access to modern technology.

Input Cost (INPUT)

Input cost is proxied by the import value of pesticides, measured in thousands of U.S. dollars, and denoted as INPUT. This variable reflects the cost burden faced by farmers when procuring essential inputs for production. Data for this variable were sourced from FAOSTAT.

Input costs directly affect farmers' production decisions and output levels. A priori, a negative relationship is expected between input cost and agricultural output.

Gross Domestic Product per Capita (GDPPC)

GDP per capita (GDPPC) represents the average economic output per person and serves as a proxy for income levels and consumer demand. It is measured in constant 2015 U.S. dollars and sourced from WDI. Higher GDP per capita is associated with improved living standards and greater investment capacity both in agriculture and in consumer goods. Rising incomes may also lead to increased demand for food and better technologies among producers.

Thus, a positive relationship is expected between GDP per capita and agricultural output. Similar studies, including that of Alemu & Adesina (2021), have documented a significant positive effect of per capita income on sectoral output.

Credit to Agriculture (CREDIT)

The CREDIT variable reflects the proportion of national credit extended to the agricultural sector, sourced from FAOSTAT. It encompasses loans provided to producers in agriculture, forestry, and fishing, including cooperatives and agribusinesses. Access to affordable and timely credit enables farmers to invest in seeds, fertilizers, technology, and infrastructure. Therefore, the expected relationship is positive.

Table 3.1. Variable Definitions

Variable	Variable Name	Unit	Source	Expected Sign
AGRO	Agricultural Output	USD, 2015 prices	FAOSTAT	
F_INF	Food Inflation	Percent (%)	GSS, ISSER	+
NF_INF	Non-Food Inflation	Percent (%)	GSS, ISSER	-
EXC	Exchange Rate	GHS/USD	FAOSTAT	-
INPUT	Input Cost	1000 USD	FAOSTAT	-
GDPPC	GDP Per Capita	Constant 2015 USD	WORLD BANK	+
CREDIT	Total Credit to Agriculture	Constant 2015 USD	FAOSTAT	+

3.6. Study Area

The research focuses on agricultural regions in Ghana with high production volumes, ensuring diversity in crop and livestock systems.

CHAPTER FOUR

RESULTS AND DISCUSSIONS

4.1. Introduction

This chapter presents and discusses the results of the study which investigates the effects of food inflation and non-food inflation on agricultural output in Ghana. The empirical analysis is based on time series data from 1991 to 2022 and employs the Autoregressive Distributed Lag (ARDL) model framework due to the mixture of stationarity levels observed among the variables. The chapter is structured to include Trend analysis, descriptive statistics, pre-diagnostic tests, estimation of both short-run and long-run ARDL models, and post-estimation diagnostic and stability tests. The interpretations are backed by economic theory and empirical expectations.

4.2. Trend analysis

Trend analysis was conducted to examine the movement of agricultural output, food inflation, and non-food inflation in Ghana over the period 1991 to 2022. The purpose was to identify the direction of change whether upward or downward and to quantify the rate of change in these variables over time. A linear regression model was used to estimate the slope coefficient of each variable against time. This helps to determine whether each variable exhibits a rising or declining trend over the years.

Descriptive statistics

The descriptive statistics provide a summary of the characteristics of the dataset used in the analysis. Agricultural output recorded a mean value of 98.65 and a standard deviation of 22.46, which suggests moderate dispersion and relative stability across the period under study. Food

inflation had a mean of 15.94 and a high standard deviation of 13.48, resulting in a coefficient of variation of 84.56%. This indicates high volatility in food prices over time. Non-food inflation had a mean of 21.06 with a coefficient of variation of 58.45%, also showing substantial variability. Among all the variables, the exchange rate was the most volatile with a coefficient of variation of 111.27%, signifying large fluctuations in the value of the Ghanaian cedi.

Input cost, GDP per capita, and credit to the agricultural sector also recorded high coefficients of variation all exceeding 70%, suggesting significant fluctuations in economic and financial conditions that may influence agricultural performance. These statistics justify the need for a dynamic econometric approach to capture the evolving relationship among these variables.

Table 4.1. Summary Statistics.

Variable	Mean	Maximum	Minimum	Std. Dev.	CV (%)
Agricultural Output (USD)	98.64875	133.648	51.46112	22.45921	22.76
Food inflation (%)	15.93946	61.23333	2.377121	13.47912	84.56
Non-food inflation (%)	21.0587	57.18333	7.508333	12.30576	58.45
Exchange rate (GHS/USD)	1.914511	8.2724	0.0367633	2.13054	111.27
Input Cost (USD)	132193.2	370868.2	14254.71	116582.6	88.17
GDP Per capita (USD)	1348.435	2066.173	878.0884	404.8899	30.03
Credit to Agriculture (USD)	3883.547	8340	270	3076.168	79.24

Note: Coefficient of Variation (CV) is calculated as CV = (σ/X) x100 where σ is the estimated standard deviation and X is the estimated mean. Source: Authors' estimation using STATA 17

4.2.1. Trend in Ghana's agricultural output from 1991 to 2022

From the estimation, the coefficient on the year variable was 1.087 with a p-value of 0.009, which is statistically significant at the 1% level. This implies a positive and statistically significant trend in agricultural output over the 32-year period. The R-squared value of 0.2062 suggests that time explains about 20.62% of the variation in agricultural output. The estimated model is as follows:

$$AGRIC.Output_t = -2082.89 + 1.087T$$

This indicates that, on average, agricultural output increased by approximately 1.09 units annually between 1991 and 2022. Although the R-squared value is modest, the positive trend is consistent with national policy efforts to improve agriculture through initiatives such as Planting for Food and Jobs and subsidy reforms in agricultural inputs. The positive direction aligns with theoretical expectations and findings in studies such as Mohammed et al. (2020) and Obe-Nwaka & Agbenyi (2023), which showed that agricultural output in Ghana has followed a long-term upward trend due to sectoral reforms and increasing investments



Figure 4.1. Agricultural output in Ghana from 1991 to 2022

Source: Authors' construct, STATA 17

4.2.2. Trend in Ghana's food inflation from 1991 to 2022

The coefficient for the year variable was -0.546 with a p-value of 0.032, statistically significant at the 5% level. The negative coefficient implies that food inflation has experienced a downward trend over the 32-year period. The adjusted R-squared value of 11.6% shows that a modest portion of the variation in food inflation is explained by time. The regression equation is:

$$\textbf{Foodinflation}_t = 1112.33 - 0.546T$$

This suggests that, on average, food inflation decreased by 0.55 percentage points per year between 1991 and 2022. This trend may be attributed to structural adjustments, macroeconomic reforms, improved food supply systems, and inflation targeting policies adopted during the Economic Recovery Programme and subsequent reforms. These findings are consistent with evidence from Alper et al., (2016), who reported inflation moderation trends in selected Sub-Saharan African economies due to policy stabilization efforts.

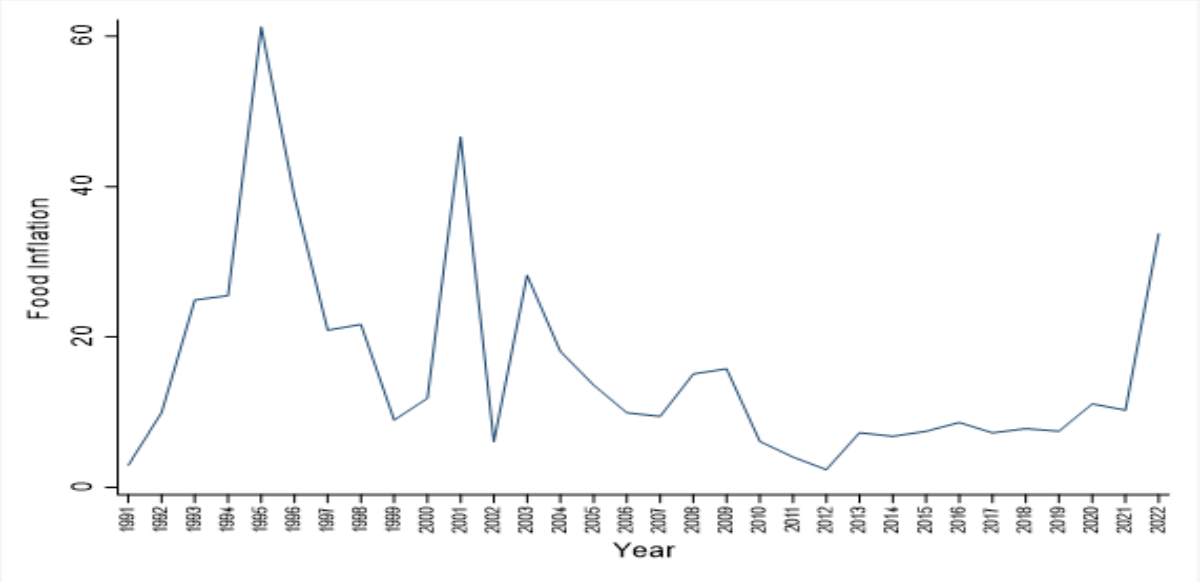


Figure 4.2. Food inflation in Ghana from 1991 to 2022

Source: Authors' construct, STATA 17

4.2.3. Trend in Ghana's non-food inflation from 1991 to 2022

The estimated coefficient for the year variable is -0.566, with a p-value of 0.014, which is statistically significant at the 5% level. The R-squared value of 18.65% indicates that time accounts for a fair portion of the variance in non-food inflation. The estimated regression equation is:

$$\textbf{Non-foodinflation}_t = 1157.70 - 0.566T$$

The negative coefficient suggests that non-food inflation declined by approximately 0.57 percentage points per year between 1991 and 2022. This is likely due to import liberalization, exchange rate stabilization, and relatively more predictable pricing in the non-food sector, particularly in consumer durables, transportation, and services. The declining trend in non-food inflation aligns with the observations of Seidu and Vasilev (2024), who found a significant reduction in Ghana's inflation over time as a result of macro-fiscal consolidation and monetary policy tightening.

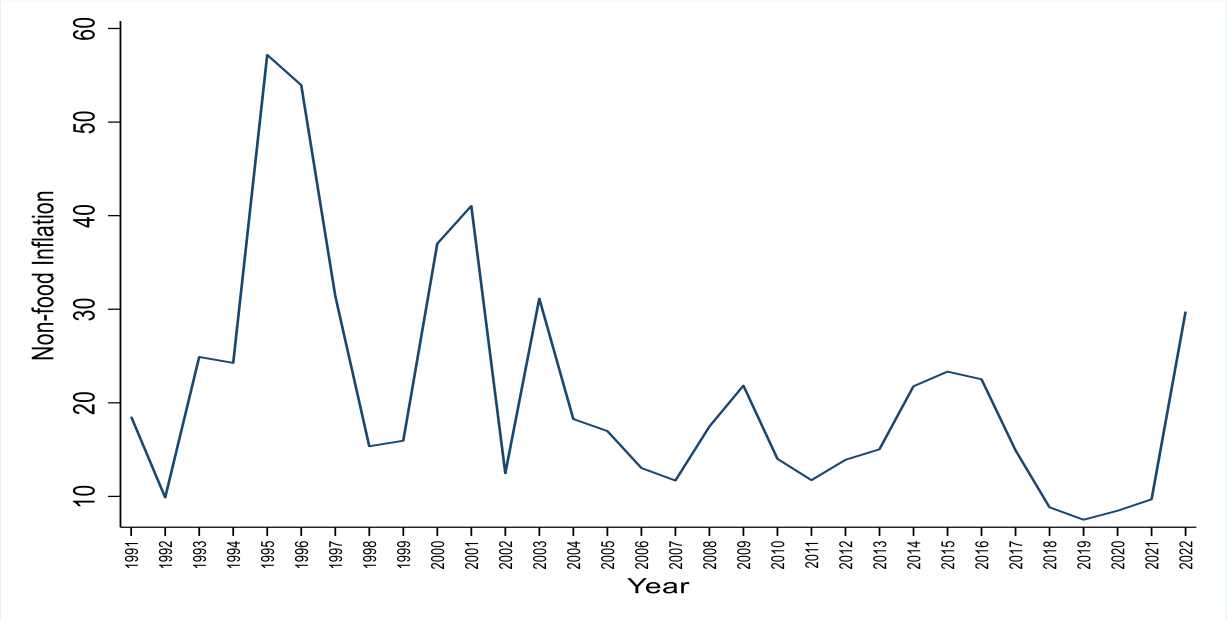


Figure 4.3. Non-food inflation in Ghana from 1991 to 2022

Source: Authors' construct, STATA 17

4.3. Magnitude and effects of food inflation and non-food inflation on Ghana's agricultural output.

Results for linear regression Model

A multiple linear regression model was first estimated as a baseline to examine the immediate linear relationships among the variables. The results showed that exchange rate had a statistically significant negative effect on agricultural output at the 1% level, suggesting that depreciation in the exchange rate increases the cost of imported agricultural inputs and thus reduces output. Input cost was found to have a positive and statistically significant effect on output at the 1% level, possibly indicating that increased input spending reflects higher investment in agriculture. GDP per capita was positively associated with output and significant at the 10% level, implying that general improvements in income levels may support agricultural performance. However, food inflation, non-food inflation, and interest rate were not statistically significant in this preliminary regression, indicating the need for a more dynamic model.

Table 4.2. Estimated Coefficients using the Linear Regression Model Approach. Dependent variable: Agricultural Output

Variable	Coefficient	Std. err.	t-value	Prob.
Food inflation	0.254	0.230	1.100	0.280
Non-food inflation	-0.182	0.234	-0.780	0.443
Ln (Exchange rate)	-35.906 ***	4.915	-7.310	0.000
Ln (Input Cost)	22.259 ***	3.413	6.520	0.000
Ln (GDP per Capita)	36.008 **	16.677	2.160	0.041
Ln (Total Credit to Agriculture)	25.700 ***	7.443	3.450	0.002
Constant	-614.993 ***	113.004	-5.440	0.000
R-squared	0.8874	**Heteroscedasticity**	0.6404	
Adjusted R-squared	0.8603	**Normality**	0.8103	
Breusch-Godfrey LM test	0.3726	**Correlogram statistic**	0.6018	

Note: ***, **, and * indicate 1%, 5% and 10% significance levels respectively.

Result for unit root/stationarity analysis

The Augmented Dickey-Fuller (ADF) test was conducted to determine the stationarity properties of each variable. The results indicated that food and non-food inflation were stationary at level, meaning they are integrated of order zero, I (0). In contrast, variables such as agricultural output, exchange rate, input cost, GDP per capita, interest rate, and credit to agriculture were not stationary at level but became stationary after first differencing, thus integrated of order one, I (1). As no variable was integrated at order two, the use of the Autoregressive Distributed Lag (ARDL) modelling technique was justified.

Table 4.3. Augmented Dickey-Fuller and Phillips Perron Test Results

Variables	ADF TEST		PP TEST	
	Level I (0)	First difference I (1)	Level I (0)	First difference I (1)
Agricultural Output (USD)	-1.639	-4.837***	-1.800	-4.812***
Food inflation (%)	-3.056**	-8.096***	-3.665***	-8.375***
Non-food inflation (%)	-3.056**	-5.743***	-3.008**	-5.868***
Ln (EXCHANGE RATE)	-2.354	-3.866***	-2.326	-3.853***
Ln (INPUTCOST)	-0.720	-4.786***	-0.771	-4.791***
Ln (GDPPC)	0.764	-3.606 ***	0.511	-3.589***
Ln (CREDIT TO AGRIC)	-2.425	-3.610***	-2.174	-3.700***

Note: *** and ** indicate 1% and 5% significance levels, respectively. Source: Authors' estimations, STATA 17

Results for ARDL bounds test for cointegration

Table 4.4. Lag-order Selection Criteria.

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-353.328				63.8752	24.0219	24.1265	24.3488
1	-138.673	429.31	49	0.000	0.00112	12.9782	13.8149	15.5938
2	-51.7736	173.8*	49	0.000	0.000163*	10.4516*	12.0205*	15.3558*

Note: *optimal lag. Source: Authors' estimation, STATA 17

The bounds testing approach to cointegration was implemented to assess the existence of a long-run relationship among the variables. The computed F-statistic was 19.819, which exceeds the upper bound critical value of 4.43 at the 1% level. This provides strong evidence of a long-run equilibrium relationship between agricultural output and the explanatory variables.

Table 4.5. ARDL bounds test result for cointegration.

Test Statistic	Value	Level of signifi-cance	Lower bound I (0)	Upper bound I (1)	Decision
F-statistic	19.819	10%	2.12	3.23	Existence of
K	6	5%	2.45	3.61	long-run relation-
		1%	3.15	4.43	ship

Source: Authors' estimation, STATA 17

Long-Run ARDL Estimates

The long-run ARDL model reveals that several macroeconomic variables exert significant influence on agricultural output. In particular, food inflation was found to have a positive and statistically significant effect on agricultural output at the 1% level, with a coefficient of 1.222. This implies that a one percent increase in food inflation leads to a 1.22 unit rise in real agricultural output. This finding is consistent with the study by Headey and Hirvonen, (2023), which also found that moderate levels of food inflation can stimulate agricultural producers by raising farmgate prices and improving profit margins, thereby incentivizing production.

Conversely, non-food inflation exerts a negative and significant influence on agricultural output at 5% level in the long-run, with a coefficient of -0.445. This suggests that rising non-food prices which include housing, energy, and transport raise the cost of living and agricultural inputs (indirectly), reducing farmers' real purchasing power and output efficiency. This supports the findings from Tenaw and Demeke, (2020) on Ethiopia, where general inflation especially non-food related was shown to undermine agricultural productivity due to rising real input costs.

The exchange rate, captured in logarithmic form, exhibited a negative and highly significant relationship with agricultural output in the long run, as evidenced by a coefficient of - 46.03. This indicates that a depreciation of the Ghanaian cedi, holding other factors constant, significantly reduces agricultural output. The implication is that Ghanaian agriculture, which depends substantially on imported inputs like fertilizers and machinery, suffers when the local currency weakens. This is aligned with findings from Iddrisu et al., (2015) who also reported a similar negative effect of exchange rate depreciation on Ghana's agricultural performance in the long-run.

The coefficient of input cost was found to be 14.83, and it is statistically significant at the 1% level. This result indicates that a 1% increase in input cost is associated with 14.83 units

increase in agricultural output, holding all other factors constant. This finding contradicts the a priori expectation, which predicted a negative relationship between input costs and output, on the assumption that higher costs of inputs such as fertilizer, pesticides, and equipment would constrain farmers' ability to invest in production, especially in a context of limited access to credit. However, this unexpected positive relationship may reflect the inclusion of imported or improved inputs, which, although costly, may have contributed to higher yields and value addition (Kim, 2020).

GDP per capita, a proxy for economic development, shows a positive and statistically significant effect in the long run with a coefficient of 25.75. This implies that as the average income of citizens improves, it enhances their ability to invest in farming or purchase more food, thereby encouraging supply-side responses in the sector. This confirms the results from Mohammed et al., (2020), where rising incomes had a complementary effect on agricultural growth.

Lastly, total credit to the agricultural sector is also highly significant with a positive coefficient of 50.72. This means that increased access to credit supports capital investment, input acquisition, and risk management by farmers, which in turn enhances productivity. This finding aligns closely with those in the article by Gibogwe and Nigo, (2023), which emphasized that the availability of financial resources significantly influences agricultural expansion in Sub-Saharan Africa.

Table 4.6. Estimated Long-Run Coefficients using the ARDL Approach. Dependent variable: Agricultural Output

Variable	Coefficient (1,2,2,2,1,2,1)	Std. Error	t-value	Prob.
Food inflation	1.222 ***	0.188	6.490	0.000
Non-food inflation	-0.445 **	0.171	-2.600	0.023
Ln (Exchange rate)	-46.035 ***	2.533	18.170	0.000
Ln (Input Cost)	14.833 ***	1.656	8.960	0.000
Ln (GDP per Capita)	25.748 ***	7.870	3.270	0.007
Ln (Total Credit to Agriculture)	50.722 ***	3.737	13.570	0.000
Constant	-607.383 ***	96.162	-6.320	0.000
R-squared	0.9861	**Heteroscedasticity**	0.6576	
Adjusted R-squared	0.9664	**Normality**	0.0532	
Breusch-Godfrey LM test	0.0043	**Correlogram statistic**	0.0810	

Note: ***, **, and * indicate 1%, 5% and 10% significance levels respectively.

Results of the error correction model

The short-run results from the ARDL model provided insights into the immediate and lagged effects of the independent variables on agricultural output. In the short run, the coefficient of food inflation at level is negative and statistically significant at 1% level, with a value of -0.7572. This suggests that a 1% increase in food inflation in the current period reduces agricultural output by approximately by 0.76 units. The lagged difference also shows a negative

and significant effect of - 0.4501, indicating that the adverse impact of food inflation persists over time. These results affirm that rising food prices reduce the purchasing power of farmers and consumers alike, potentially disrupting food input markets and lowering output.

Conversely, non-food inflation shows an unexpected but positive and significant effect on agricultural output in both the current and lagged periods. The coefficient of $\text{Non} - \text{Food inflation}_t$ is 0.4534, and that of $\text{Non} - \text{Food inflation}_{t-1}$ is 0.5672, both statistically significant at 1%. These results suggest that non-food inflation may indirectly stimulate agricultural production in the short term, possibly by reallocating consumption toward locally produced food or incentivizing investment in agriculture as a hedge against rising non-food prices.

The exchange rate has a strong negative impact in both the current and lagged periods. The coefficient for $\text{Ln (Exchange rate)}_t$ is -43.22, and $\text{Ln (Exchange rate)}_{t-1}$ is -35.59, both highly significant at the 1% level. These results show that a depreciation of the Ghanaian cedi relative to the U.S. dollar reduces agricultural output in the short term too. This is likely due to the increased cost of imported agricultural inputs such as fertilizers, machinery, and pesticides. These findings align with those of Obe-Nwaka and Agbenyi (2018), who observed that exchange rate volatility negatively affects agricultural output in Nigeria in the short-run.

Input cost had a negative short-run effect in its first-differenced form, with a coefficient of -12.31, and it is statistically significant at the 1% level. This aligns with the a priori expectation, confirming that an increase in input cost in the short term reduces agricultural output. This may be because farmers, particularly smallholders who dominate Ghana's agriculture, do not respond immediately to rising costs by increasing production. Instead, they may reduce the scale of operations, use fewer inputs, or delay planting decisions due to financial constraints. This observation is consistent with the findings of Baquedano et al.,(2022) who also reported

that higher input prices constrain agricultural productivity due to limited affordability and access among rural farmers.

The short-run coefficients of GDP per capita are negative but statistically insignificant. The current period effect is -18.77, and the lagged effect is -38.19, both of which lack statistical significance. While this contradicts the expectation that improved economic growth should enhance agricultural productivity through better infrastructure and income levels, the insignificance may be due to time lags in policy translation or structural imbalances between growth sectors in Ghana. This disconnect between economic growth and agriculture has also been noted in the work of Atchadé and Nougbodé, (2024), who found a unidirectional causality from agriculture to GDP, but not vice versa.

Finally, the short-run effect of total credit to agriculture is negative and statistically significant at the 1% level (-21.58). This result contradicts the a priori expectation that increased credit should boost agricultural output. A plausible explanation could be the inefficiency in the disbursement of agricultural credit or diversion of funds for non-productive uses. This concern was also raised by Prah et al., (2025) who found that agricultural credit in Ghana often lacks proper targeting, reducing its effectiveness.

The coefficient of the error correction term (ECT) was estimated at -0.9411, and it is statistically significant at the 1% level, indicating a stable and converging system. It suggests that approximately 94.1% of the disequilibrium from the previous year's shock to agricultural output is corrected in the current period, implying a very rapid speed of adjustment toward long-run equilibrium.

Table 4.7. Results of the ARDL Error Correction Model. Dependent variable: ΔAgricultural Output

Variables	Coefficient	Std. Error	t-value	Prob.
$\Delta\textbf{Food inflation}_{t}$	-0.757 ***	0.142	-5.35	0.000
$\Delta\textbf{Food inflation}_{t-1}$	-0.450 ***	0.099	-4.54	0.001
$\Delta\textbf{Non} - \textbf{Food inflation}_{t}$	0.453 ***	0.128	3.53	0.004
$\Delta\textbf{Non} - \textbf{Food inflation}_{t-1}$	0.567 ***	0.112	5.07	0.000
$\Delta\textbf{Ln}\ (\textbf{Exchange rate})_{t}$	-43.222 ***	7.092	-6.09	0.000
$\Delta\textbf{Ln}\ (\textbf{Exchange rate})_{t-1}$	-35.591 ***	8.347	-4.26	0.001
$\Delta\textbf{Ln}\ (\textbf{Input Cost})_{t}$	-12.306 ***	2.326	-5.29	0.000
$\Delta\textbf{Ln}\ (\textbf{GDP per capita})_{t}$	-18.765	29.127	-0.64	0.532
$\Delta\textbf{Ln}\ (\textbf{GDP per capita})_{t-1}$	-38.192	23.365	-1.63	0.128
$\Delta\textbf{Ln}\ (\textbf{Credit to Agriculture})_{t}$	-21.579 ***	6.777	-3.18	0.008
ECT (-1)	-0.941 ***	0.103	-9.15	0.000
Constant	-607.383 ***	96.162	-6.32	0.000

R-squared	0.9861	**Heteroscedasticity**	0.6576
Adjusted R-squared	0.9664	**Normality**	0.0532
Breusch-Godfrey LM test	0.0043	**Correlogram statistic**	0.0810

Note: *** and ** indicate 1% and 5% significance levels, respectively. Source: Author's estimation, STATA 17

Post-diagnostic test results

Following the estimation of the ARDL model and the associated Error Correction Model (ECM), a series of post-estimation diagnostic tests were conducted to evaluate the validity, consistency, and robustness of the model. These diagnostic tests assessed the presence of autocorrelation, heteroskedasticity, normality of residuals, and model specification error. The tests are crucial in ensuring that the assumptions underlying the classical linear regression model are satisfied and that the results can be reliably used for policy and academic inferences (Kumar, 2023).

Test for Normality of Residuals

To assess whether the residuals follow a normal distribution, the Skewness/Kurtosis (sktest) test was used. The null hypothesis states that the residuals are normally distributed. The test results indicate a joint chi-square statistic of 5.87 with a p-value of 0.0532, which is marginally above the 5% significance level. This implies that we fail to reject the null hypothesis at the 5% level, although the distribution slightly deviates from normality. In practice, this marginal non-normality is not considered severe enough to invalidate the model's inferences, especially given the sample size and the robustness of ARDL estimators under mild departures from normality.

Test for Heteroskedasticity

To verify whether the variance of the residuals is constant, the Breusch–Pagan/Cook–Weisberg test for heteroskedasticity was used. The null hypothesis assumes homoscedasticity (i.e., constant variance of residuals). The test produced a Chi-square statistic of 0.20 with a p-value

of 0.6576, which is well above the 5% significance threshold. Therefore, we fail to reject the null hypothesis, indicating that the residuals are homoscedastic and that the model does not suffer from heteroskedasticity. This supports the efficiency of the OLS and ARDL estimates.

Test for Autocorrelation

To test for serial correlation among the residuals of the model, the Breusch–Godfrey LM test was employed. The null hypothesis of the test states that there is no autocorrelation in the residuals. The test was conducted with one lag as recommended by the model's lag structure. The results of the test yielded a Chi-square value of 8.148 with a corresponding p-value of 0.0043, which is significant at the 1% level. Consequently, the null hypothesis of no autocorrelation is rejected. This indicates the presence of first-order serial correlation in the residuals of the model, suggesting that some important dynamics may not have been fully captured, or that more lags may need to be included.

To further confirm this finding, the Portmanteau Q-test (Correlogram test) was performed. The test result showed a Q-statistic of 19.32 with a p-value of 0.0810 for 12 lags. This test failed to reject the null hypothesis of no autocorrelation at the 5% level, implying that autocorrelation is not a significant concern when considering the entire lag structure. Taken together, the results suggest mild serial correlation that may warrant robustness checks but does not fundamentally undermine the model.

Test for Model Specification Error

The Ramsey Regression Equation Specification Error Test (RESET) was conducted to determine whether the model suffers from omitted variable bias or functional form misspecification. The null hypothesis posits that the model is correctly specified.

The test yielded an F-statistic of 0.81 with a p-value of 0.5175, which is not statistically significant at any conventional level. As such, the null hypothesis cannot be rejected, suggesting that the model is correctly specified and that there is no evidence of omitted variables or incorrect functional form. This reinforces the credibility of the regression specification.

Table 4.8. Post-diagnostic test results

Test	Test method	Prob. > Chi2, Prob.>F & Prob.>Q
Normality	Skewness & Kurtosis	0.0532
Heteroscedasticity	Breusch-Pagan-Godfrey	0.6576
Serial correlation	Breusch Godfrey LM	0.0043
Serial correlation	Correlogram (Prob>Q)	0.081
Model specification error	Ramsey RESET test	0.5175

Source: Authors' estimations, STATA 17

Stability test

The graphical presentation from the CUSUM and CUSUMSQ plots revealed that the recursive residuals remained well within the 5% confidence interval across the entire study period from 1991 to 2022. This indicates that the ARDL model is structurally stable and free from significant parameter shifts over time.

The implication is that the estimated relationships between agricultural output and its macroeconomic determinants particularly food and non-food inflation are not sensitive to short-term disturbances or regime changes in Ghana's economic policy framework. This enhances the reliability of the model for both in-sample interpretation and policy analysis.

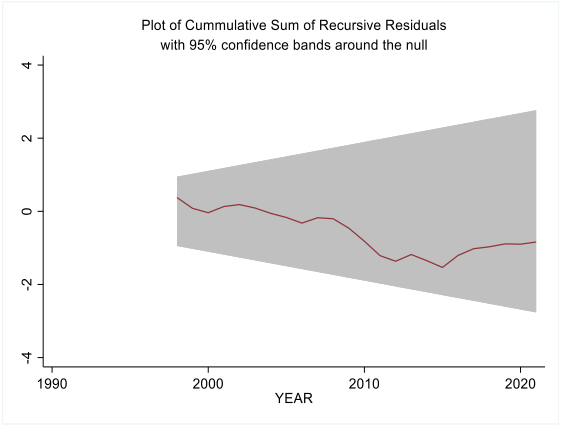


Fig.4.4. Stability test (CUSUM).
Source: Authors' estimation, STATA 17

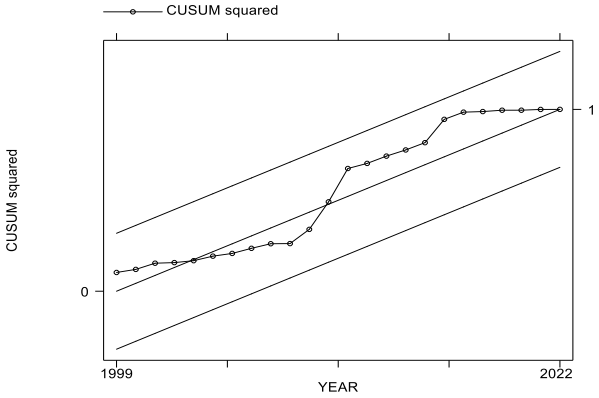


Fig.4.5. Stability test (CUSUMSQ).
Source: Authors' estimation, STATA 17

CHAPTER FIVE

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

5.1. Introduction

This chapter presents the summary of major findings, conclusion, and policy recommendations derived from the econometric investigation of the effect of food and non-food inflation on agricultural output in Ghana. The chapter begins with a concise summary of key insights obtained through descriptive and econometric analysis and is followed by conclusions and actionable policy suggestions informed by the findings.

5.2. Summary of Findings

This study investigated the effect of food and non-food inflation on agricultural output in Ghana using annual time series data from 1991 to 2022. Trend analysis showed that agricultural output has experienced a consistent upward trend over the study period, while both food and non-food inflation showed statistically significant downward trends, reflecting improved macroeconomic stability.

The study also estimated a linear regression, an ARDL model, and an Error Correction Model (ECM) to determine the long-run and short-run effects of inflation and other macroeconomic variables on agricultural output. The results revealed a positive and significant long-run effect of food inflation, GDP per capita, input cost, and credit to agriculture on agricultural output. On the other hand, non-food inflation had a negative long-run effect, aligning with its cost-push implications.

In the short run, food inflation showed a negative effect, suggesting immediate shocks to production, while other variables such as non-food inflation showed positive but less pronounced effects.

Post-estimation diagnostic tests confirmed the validity, stability, and reliability of the estimated models. Overall, the study found that macroeconomic indicators like inflation, income levels, and credit allocation are key determinants of agricultural output in Ghana.

5.3. Conclusion of the Study

The trend analysis revealed a statistically significant upward trend in agricultural output in Ghana from 1991 to 2022. Specifically, the annual growth rate of agricultural value added was estimated at approximately 1.09 units per year, suggesting gradual but consistent improvements in agricultural performance. Conversely, both food and non-food inflation showed statistically significant downward trends over the same period, with negative coefficients indicating that inflationary pressures have declined, possibly due to improvements in Ghana's macroeconomic stability and targeted policy interventions in the past decade.

Using the OLS estimation technique and subsequently applying the ARDL model, the study established both short-run and long-run relationships between agricultural output and the explanatory variables. The long-run estimates revealed that food inflation positively and significantly influences agricultural output. This result suggests that rising food prices, while burdensome for consumers, may serve as an incentive for producers to expand output in anticipation of greater returns, as observed in Headey and Hirvonen, (2023). On the contrary, non-food inflation was found to exert a negative long-run effect, confirming the expectation that increases in general cost-of-living and input prices can erode profit margins and limit production capacity in the agricultural sector.

The exchange rate, had a negative and significant influence with agricultural output in the long run effect. Which conforms the implication that Ghanaian agriculture, which depends substantially on imported inputs like fertilizers and machinery, suffers when the local currency weakens.

In the short-run dynamics, food inflation with both current and lagged period had a negative and statistically significant impact on agricultural output. This suggests a temporal adjustment effect, where producers are initially constrained by rising food prices due to immediate cost pressures or market rigidities. Meanwhile, non-food inflation exhibited positive short-run effects, potentially reflecting strategic adjustments by producers anticipating inflationary pressures. Exchange rate still confirmed a negative and statistically significant effects on agricultural output in the short-run. The error correction term (ECT) was negative and highly significant, indicating a stable system that corrects deviations from long-run equilibrium at a reasonable speed.

Overall, the findings reinforce the complexity of inflation-output dynamics. While certain types of inflation (e.g., food price inflation) may drive production incentives in the long run, they may also impose short-term burdens that hinder immediate agricultural response. Similarly, non-food inflation and rising input costs must be carefully monitored due to their dual and time-varying effects on agricultural output.

5.4. Policy Recommendations

In light of the findings from this study, the following policy measures are proposed to enhance agricultural output while managing inflationary pressures in Ghana:

The positive long-run relationship between food inflation and agricultural output suggests that rising food prices can serve as production incentives. Policymakers should, therefore, adopt

pricing strategies that guarantee minimum returns to farmers, especially during harvest seasons. However, these strategies must be balanced with consumer welfare thus, targeted subsidies or food safety net programs should be introduced for vulnerable populations to mitigate the regressive effects of inflation.

Given the negative long-run impact of non-food inflation, government should prioritize the stabilization of general prices, especially those related to utilities, fuel, and transportation, which indirectly affect the cost of agricultural operations. This can be achieved through prudent fiscal policy, the reduction of energy tariffs for agribusinesses, and macroeconomic tools aimed at curbing unnecessary price hikes.

Although input costs were positively associated with agricultural output in the long run, the short-run effects were negative, indicating immediate constraints to production. The government should expand agricultural input subsidy programs and improve the distribution of critical inputs such as fertilizers, seeds, and pesticides. Additionally, import tariffs on essential agricultural inputs should be reviewed to enhance affordability and ensure timely availability.

The study confirmed the positive role of credit to the agricultural sector in boosting output. Therefore, strengthening the role of development banks and rural financial institutions in providing affordable credit is essential. Policies should also incentivize private banks to allocate a minimum share of their portfolio to agriculture, with mechanisms in place to de-risk lending to smallholder farmers.

Since GDP per capita was found to influence agricultural output significantly, national planning should incorporate agricultural development as a central pillar of economic growth. Investments in rural infrastructure, education, and healthcare can raise overall productivity and living standards, creating a feedback loop that benefits the agricultural sector.

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APPENDIX

Appendix A: Regression analysis of Ghana's Agricultural output

Agricultural Output	Coefficient	Std. Erorr	t-value	Prob.
YEAR	1.087234 ***	0.3894407	2.79	0.009
constant	-2082.886 **	781.4211	-2.67	0.012
R-squared	0.2062		**Prob > F**	0.009
Adjusted R-squared	0.1798		**F-test**	7.79

Note: ***, **, and * indicate 1%, 5% and 10% significance levels respectively.

Appendix B: Regression analysis of Ghana's Food inflation

Food inflation	Coefficient	Std. Erorr	t-value	Prob.
YEAR	-0.5464179 **	0.242628	-2.25	0.032
constant	1112.327 **	486.8381	2.28	0.03
R-squared	0.1446		**Prob > F**	0.032
Adjusted R-squared	0.1161		**F-test**	5.07

Note: ***, **, and * indicate 1%, 5% and 10% significance levels respectively.

Appendix C: Regression analysis of Ghana's Non-food inflation

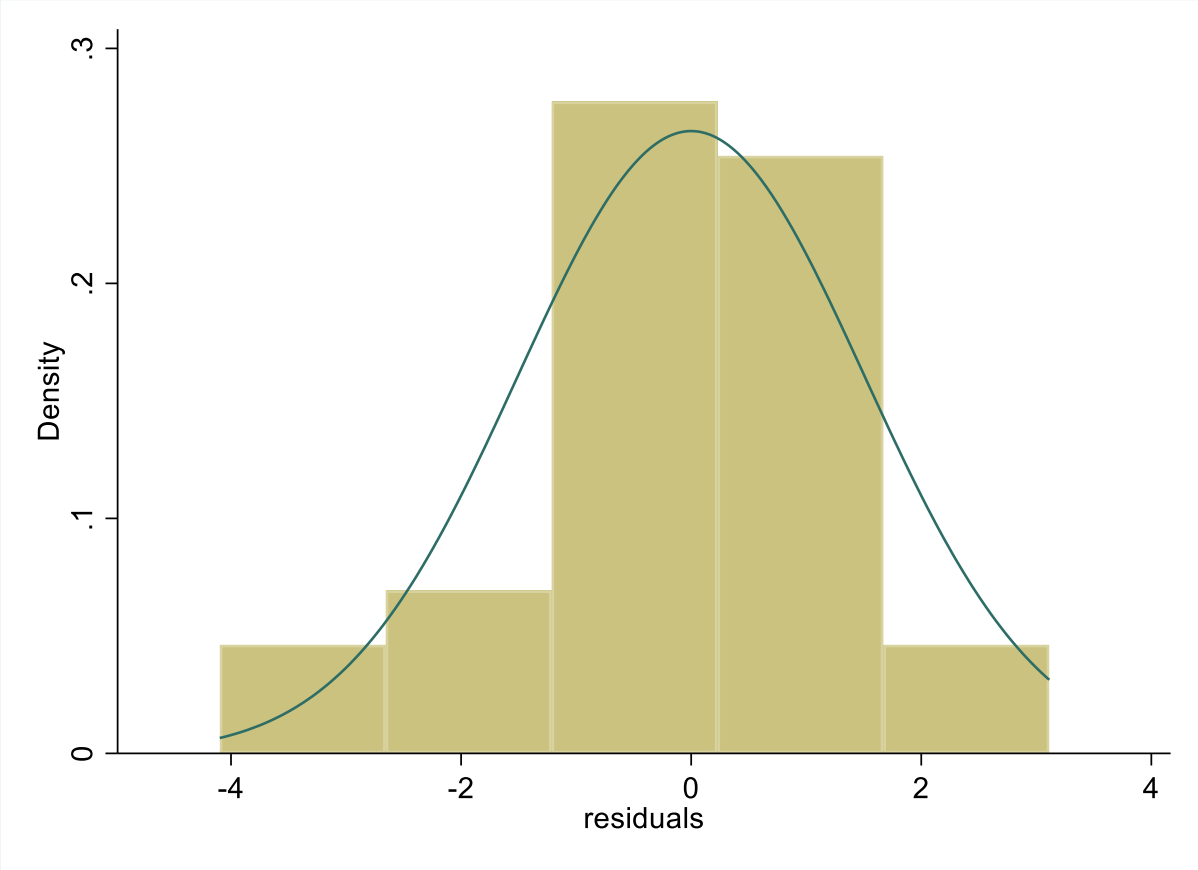
Non-food inflation	Coefficient	Std. Erorr	t-value	Prob.
YEAR	-0.5664794 **	0.2160183	-2.62	0.014
constant	1157.7 **	433.4453	2.67	0.012
R-squared	0.1865		**Prob > F**	0.014
Adjusted R-squared	0.1594		**F-test**	6.88

Note: ***, **, and * indicate 1%, 5% and 10% significance levels respectively.

Appendix D: Shapiro-Wilk W test for normal data (Normality test)

Variable	Observations	W	V	z	Prob>z
Residuals	30	0.94562	1.729	1.132	0.12889

H_0: The data follows a normal distribution.



Appendix E: Data of the study

YEAR	FOOD INFLATION (%)	NON-FOOD INFLATION (%)	EXCHANGE RATE (GHS/USD)	INPUTCOST (TOTAL PESTICIDE IMPORT VALUE, 1000 USD)	GDP PER CAPITA (CONSTANT 2015 USD)	TOTAL CREDIT TO AGRICULTURE (CONSTANT 2015, MILLION USD)	AGRICULTURAL OUTPUT (Value Added Deflator (Agriculture, forestry and fishery), USD, 2015 PRICES)
1991	2.980	18.520	0.037	15910.044	878.088	270.000	114.982
1992	9.967	9.883	0.044	15910.044	889.759	379.000	111.302
1993	24.925	24.900	0.065	16242.238	910.156	390.000	83.462
1994	25.508	24.267	0.096	16574.432	917.857	436.000	76.504
1995	61.233	57.183	0.120	16906.626	933.314	439.000	89.813
1996	38.633	53.950	0.164	17238.820	953.731	543.000	91.965
1997	20.917	31.433	0.205	28563.592	970.391	772.000	80.507
1998	21.675	15.358	0.231	14425.961	991.588	920.000	83.693
1999	8.967	15.950	0.267	14273.617	1010.252	1291.000	82.563
2000	11.867	37.008	0.545	14254.711	1021.638	1783.000	51.461
2001	46.614	41.042	0.716	20372.030	1034.582	1644.200	52.725
2002	6.084	12.467	0.792	30043.736	1051.810	1637.400	58.462
2003	28.217	31.142	0.867	48003.356	1077.029	1544.900	70.939
2004	18.101	18.267	0.899	73369.569	1107.614	1520.000	80.136
2005	13.618	16.983	0.905	69708.748	1142.600	1840.000	91.760
2006	9.931	13.033	0.915	80723.526	1184.527	2289.000	99.085
2007	9.457	11.692	0.933	110097.894	1204.566	3876.000	115.297
2008	15.095	17.467	1.058	143005.733	1281.835	4671.000	133.277
2009	15.788	21.842	1.409	164509.066	1310.813	4689.000	119.286
2010	6.124	14.017	1.431	255201.662	1380.368	4641.000	126.956
2011	4.044	11.733	1.512	370868.248	1536.860	4766.000	130.654
2012	2.377	13.908	1.796	336587.835	1639.519	5761.000	126.614
2013	7.289	15.033	1.954	241206.493	1717.526	6491.000	133.648
2014	6.804	21.767	2.900	248321.879	1725.294	7442.000	109.393
2015	7.480	23.333	3.668	246828.689	1721.700	8215.000	100.000
2016	8.637	22.517	3.910	189035.955	1739.727	8340.000	113.979
2017	7.272	14.933	4.351	194941.122	1840.299	8015.000	108.307
2018	7.828	8.833	4.587	203926.236	1914.260	8223.000	106.644
2019	7.486	7.508	5.216	154205.564	1998.308	8014.000	98.888
2020	11.114	8.467	5.596	243170.604	1968.967	7750.000	102.798
2021	10.293	9.692	5.806	332135.645	2028.784	7781.000	112.200
2022	33.736	29.750	8.272	303618.010	2066.173	7900.000	99.457